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Meta-analysis strategies in genome-wide association studies

Konstantinos Hatzikotoulas
10.11. – 14.11.2025

Agenda

1. Introduction
2. Principles of meta-analysis
3. Meta-analysis approaches
4. Considerations
5. Meta-analysis results

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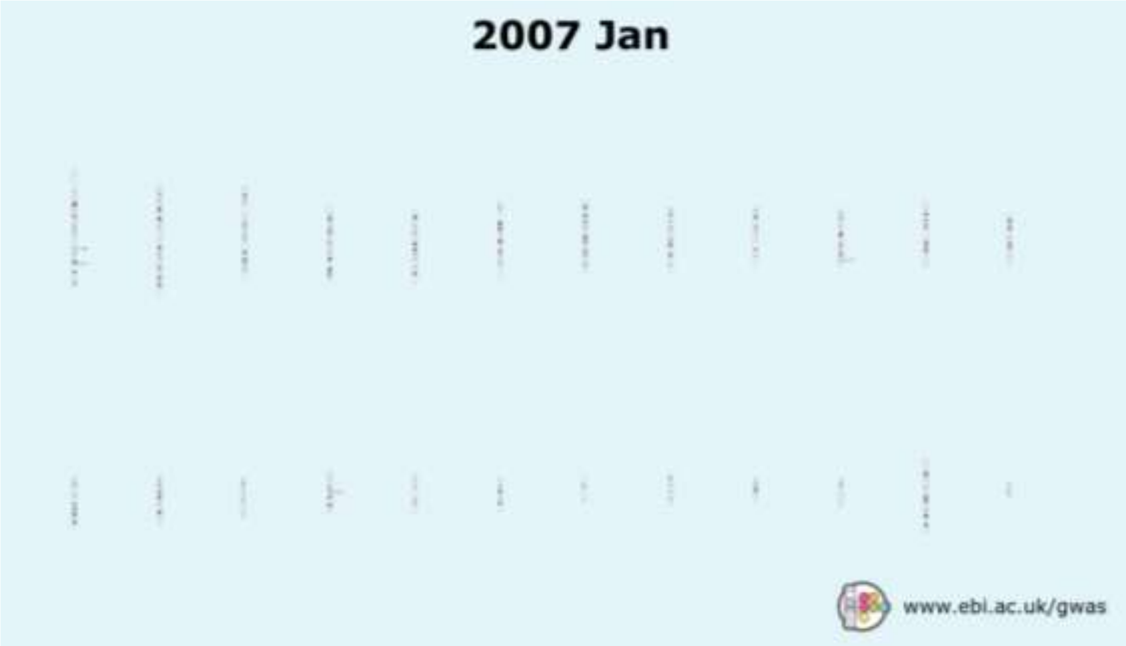
1

Introduction

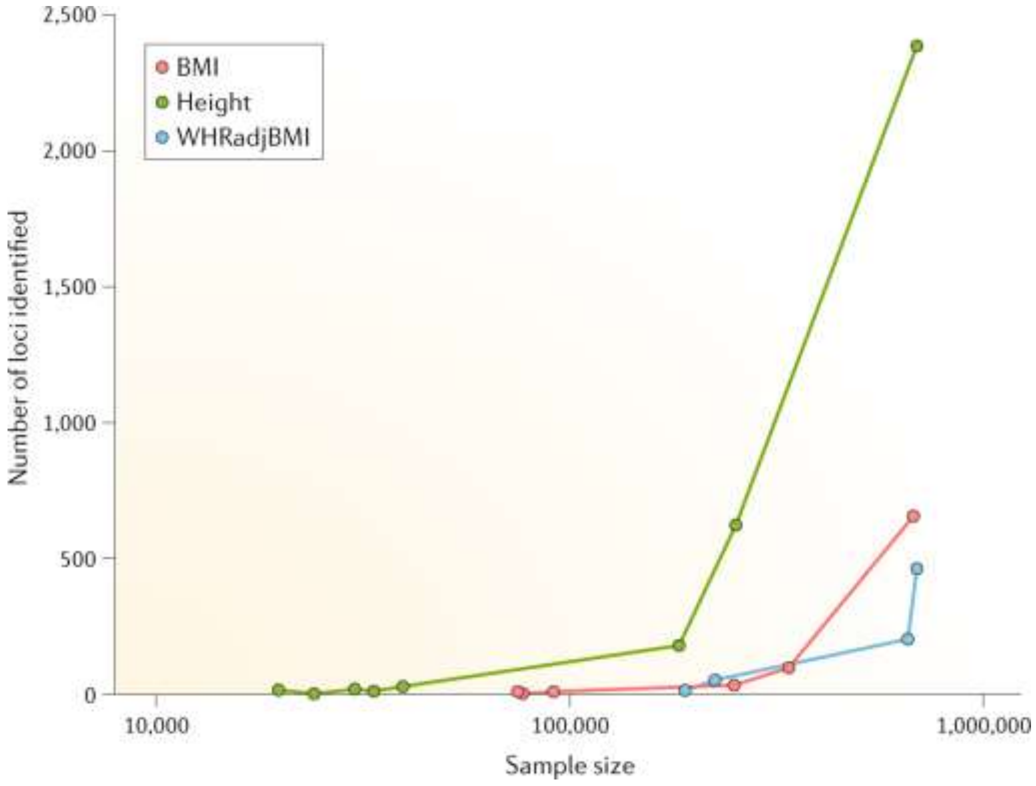
Genome-wide association studies

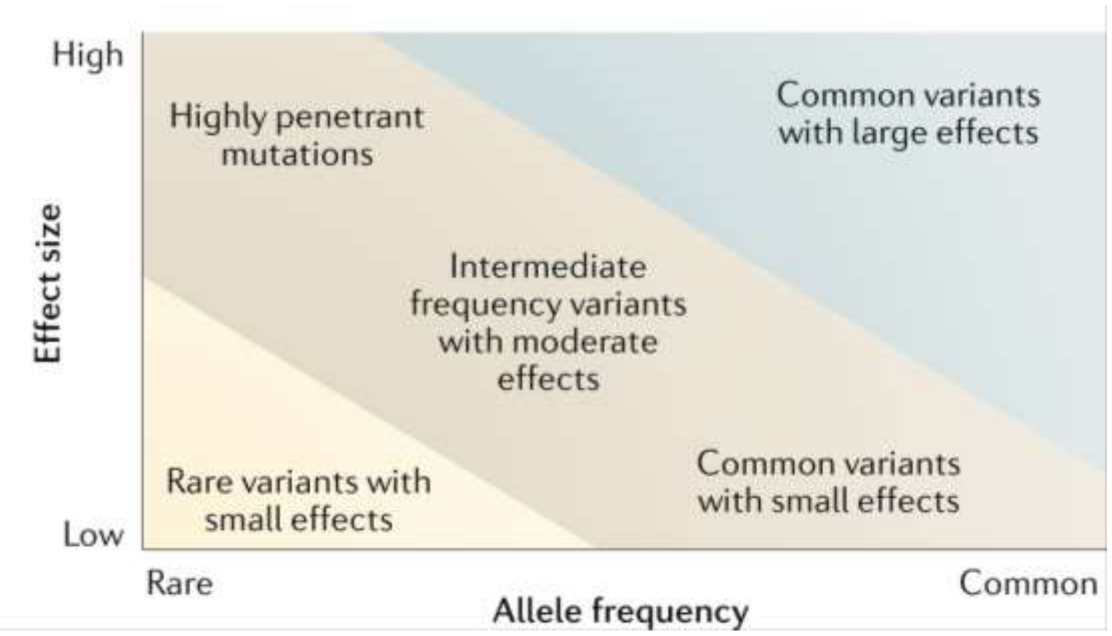


Now: https://www.ebi.ac.uk/gwas/diagram#

https://www.ebi.ac.uk/gwas/

Motivation for meta-analysis





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Tam et al. 2019. Nat Rev Genet

What we could achieve by conducting meta-analysis

- Increase the statistical power for GWAS.
- Improve the effect size estimations, which could facilitate downstream analyses. (For example, PRS or MR).
- Provide opportunities to study the less prevalent or understudied diseases and populations.
- Cross-validate findings across different studies.
- Can be carried out sequentially and updated when new GWAS from the same trait emerge.

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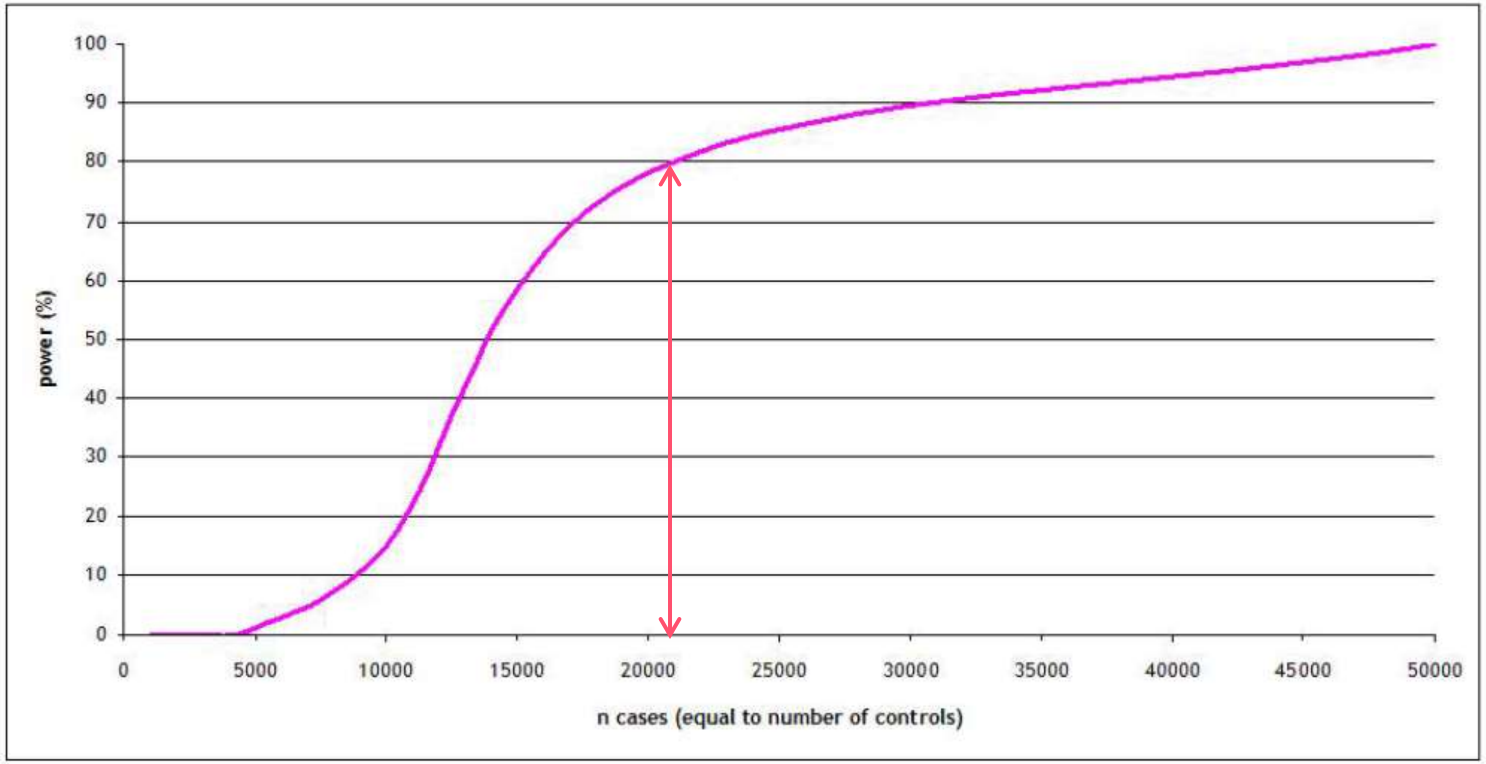
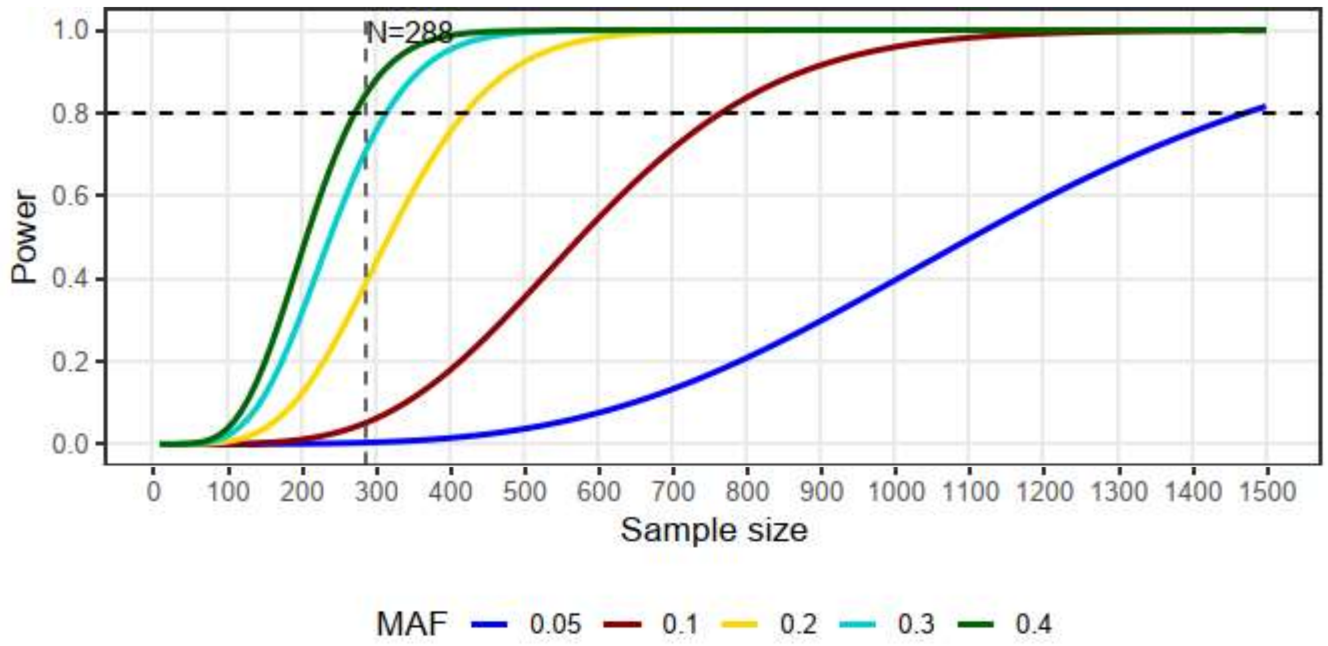
Statistical power

Power to detect association ($p = 5 \times 10^{-8}$) at a variant with risk allele frequency 0.30 and allelic OR 1.10



Statistical power

- Increased sample size → access to low-frequency variants (1-5%) and rare variants (<1%)



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Consortia

- Consortia to study specific traits of interests and gather multiple GWAS across different ancestries

https://www.genetics-osteoarthritis.com/

https://diagram-consortium.org/t2dggiabout.html

nature > articles > article

Article | Open Access | Published: 12 October 2022

A saturated map of common genetic variants associated with human height

Loic Yengo, Sailaja Vedantam, Eirini Marouli, Julia Sidorenko, Eric Bartell, Saori Sakaue, Marielisa Graff, Anders U. Eliasen, Yunxuan Jiang, Sridharan Raghavan, Jenkai Miao, Joshua D. Arias, Sarah E. Graham, Ronen E. Mukamel, Cassandra N. Spracklen, Xianyong Yin, Shyh-Huei Chen, Teresa Ferreira, Heather H. Highland, Yingjie Ji, Tugce Karaderi, Kuang Lin, Kreete Lüll, Deborah E. Malden, 23andMe Research Team, VA Million Veteran Program, DiscovEHR (DiscovEHR and MyCode Community Health Initiative), eMERGE (Electronic Medical Records and Genomics Network), Lifelines Cohort Study, The PRACTICAL Consortium, Understanding Society Scientific Group, ... Joel N. Hirschhorn + Show authors

Nature 610, 704–712 (2022) | Cite this article

45k Accesses | 2 Citations | 1350 Altmetric | Metrics

>12,000 independent SNPs

What we could achieve by conducting meta-analysis

- Can be carried out **sequentially** and updated when new GWAS from the same trait emerge

nature > nature genetics > articles > article

Article | Published: 08 October 2018

Fine-mapping type 2 diabetes loci to single-variant resolution using high-density imputation and islet-specific epigenome maps

Anubha Mahajan, Daniel Taliun, Matthias Thurner, Neil R. Robertson, Jason M. Torres, N. William Rayner, Anthony J. Payne, Valgerdur Steinthorsdottir, Robert A. Scott, Niels Grarup, James P. Cook, Ellen M. Schmidt, Matthias Wuttke, Chloé Sarnowski, Reedik Mägi, Jana Nano, Christian Gieger, Stella Trompet, Cécile Lecoeur, Michael H. Preuss, Bram Peter Prins, Xiuqing Guo, Lawrence F. Bielak, Jennifer E. Below, … Mark I. McCarthy + Show authors

32 GWAS
74,124 T2D cases,
824,006 controls

nature > nature genetics > articles > article

Article | Published: 12 May 2022

Multi-ancestry genetic study of type 2 diabetes highlights the power of diverse populations for discovery and translation

Anubha Mahajan, Cassandra N. Spracklen, Weihua Zhang, Maggie C. Y. Ng, Lauren E. Petty, Hidetoshi Kitajima, Grace Z. Yu, Sina Rueger, Leo Speidel, Young Jin Kim, Momoko Horikoshi, Josep M. Mercader, Daniel Taliun, Sanghoon Moon, Soo-Heon Kwak, Neil R. Robertson, Nigel W. Rayner, Marie Loh, Bong-Jo Kim, Joshua Chiou, Irene Miguel-Escalada, Pietro della Briotta Parolo, Kuang Lin, Fiona Bragg, FinnGen, eMERGE Consortium, … Andrew P. Morris + Show authors

122 GWAS
180,834 T2D cases,
1,159,055 controls

nature > articles > article

Article | Open access | Published: 19 February 2024

Genetic drivers of heterogeneity in type 2 diabetes pathophysiology

Ken Suzuki, Konstantinos Hatzikotoulas, Lorraine Southam, Henry J. Taylor, Xianyong Yin, Kim M. Lorenz, Ravi Mandla, Alicia Huerta-Chagoya, Giorgio E. M. Melloni, Stavroula Kanoni, Nigel W. Rayner, Ozvan Bocher, Ana Luiza Arruda, Kyuto Sonehara, Shinichi Namba, Simon S. K. Lee, Michael H. Preuss, Lauren E. Petty, Philip Schroeder, Brett Vanderwerff, Mart Kals, Fiona Bragg, Kuang Lin, Xiuqing Guo, VA Million Veteran Program, … Eleftheria Zeggini + Show authors

Nature **627**, 347–357 (2024) | Cite this article

136 GWAS
428,452 T2D cases,
2,107,149 controls

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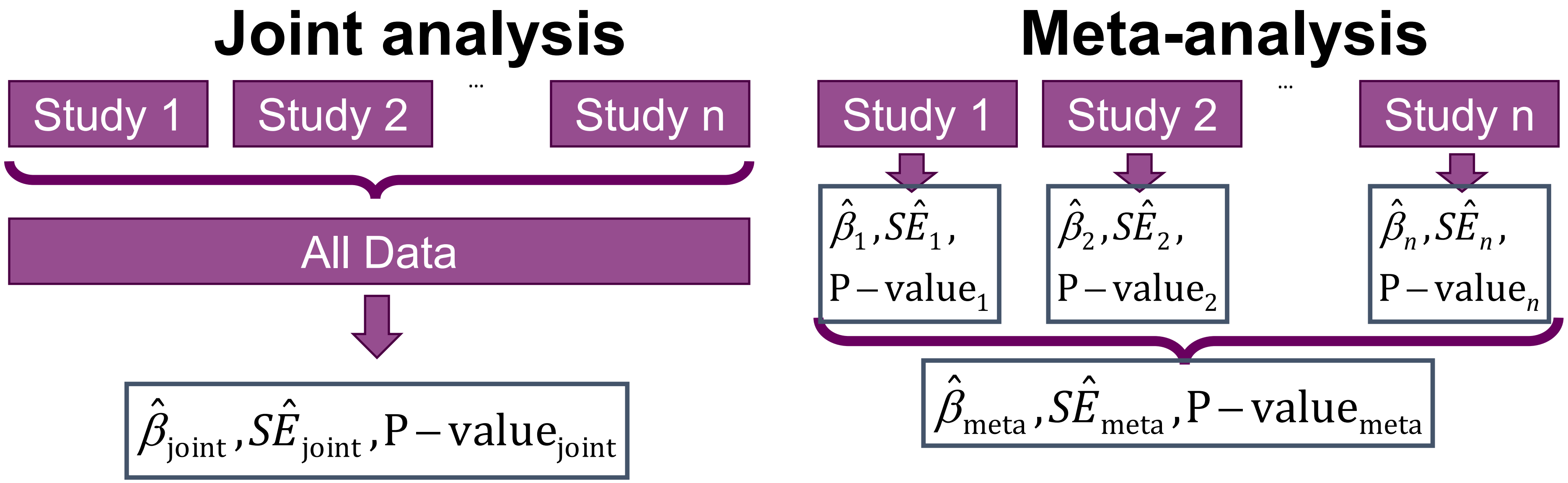
Joint (Pooled) vs Meta-analysis

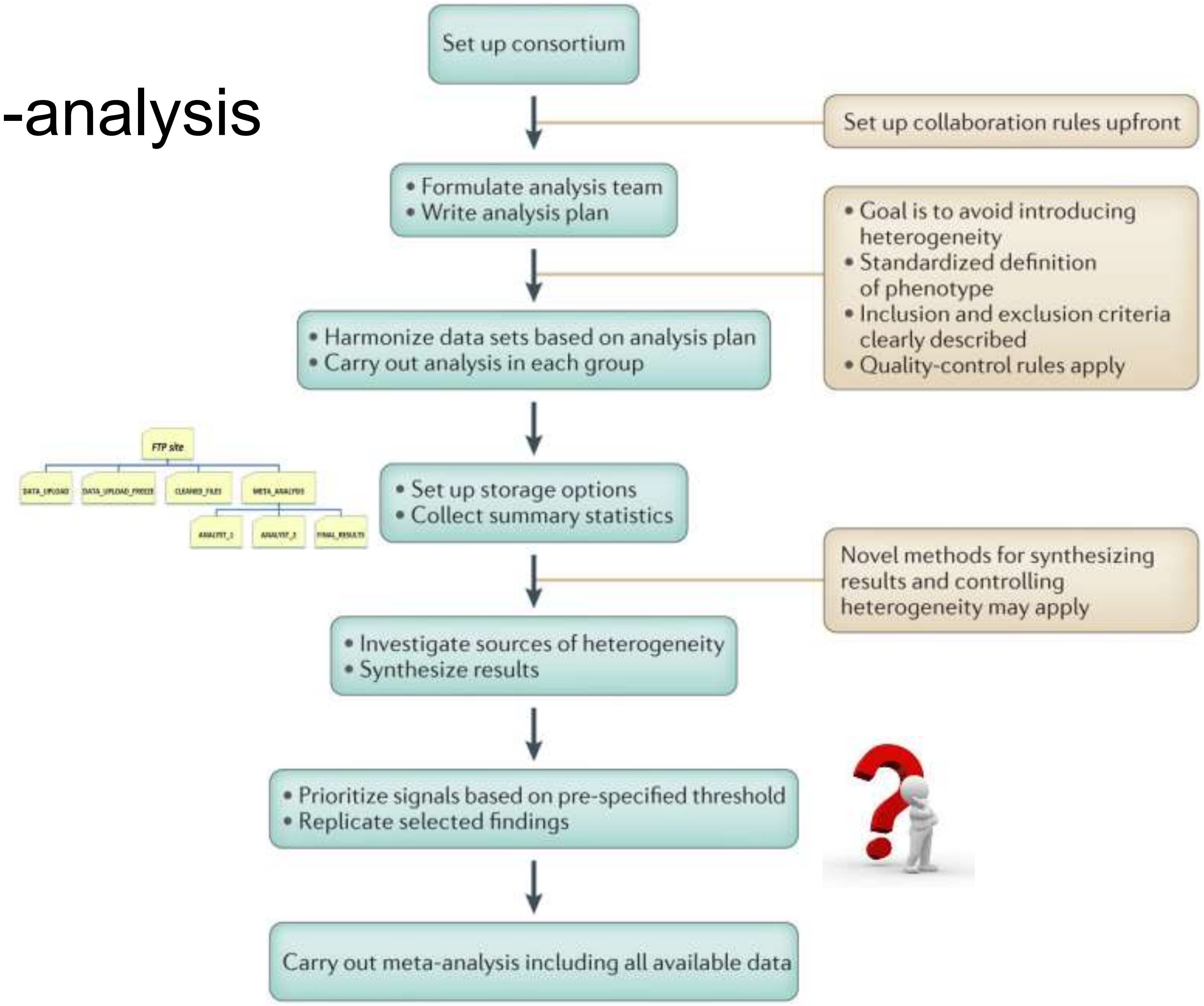


- ✓ For common variants, joint and meta-analysis have similar power (Lin & Zeng, *Genet. Epidemiol.*, 2010)
- ✓ Pooled analysis generally exhibits better statistical power while effectively adjusting for population stratification in multi-ancestry genome-wide association methods (Dias et al., *AJHG*, 2025)
- ✓ However, data sharing restrictions, such as those imposed by General Data Protection Regulation (GDPR), prevents the aggregation of individual-level data in “joint analysis”.

2

Principles of meta-analysis

Meta-analysis



Nature Reviews | Genetics

Meta-analysis pipeline

Phenotype definition

↓

QC

↓

Imputation (reference, pre- & post imputation QC)

↓

Association testing (Correction for covariates and population structure)

↓

QC prior to meta-analysis

↓

Centralized meta-analysis

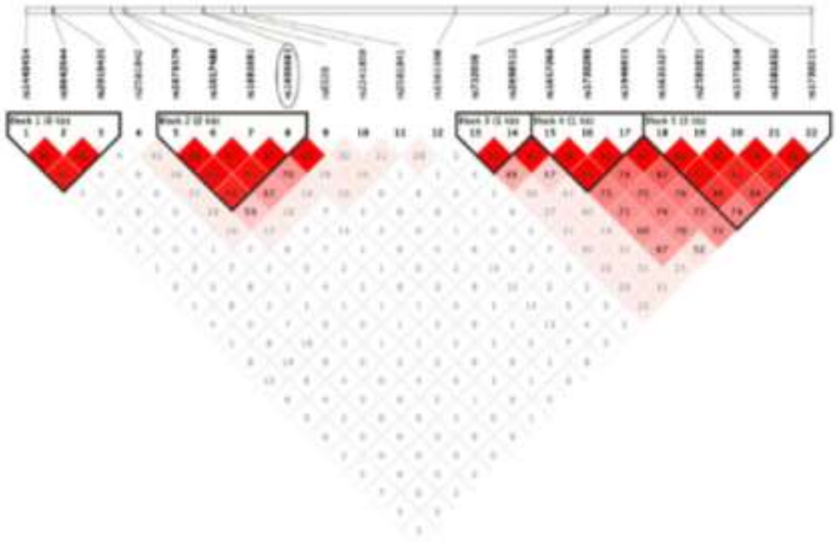
↓

Meta-analysis QC

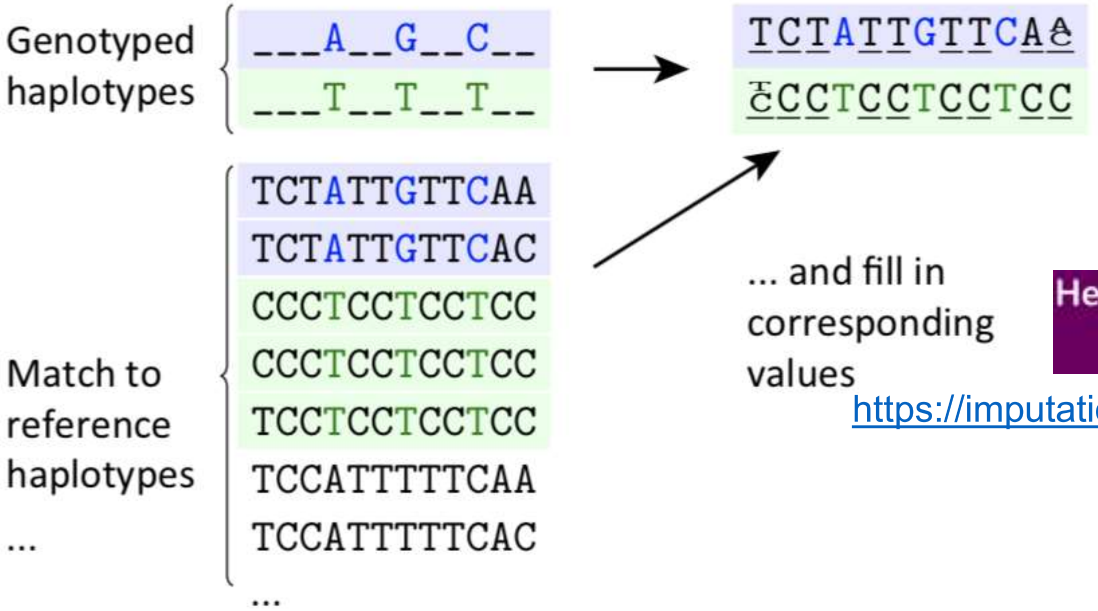
↓ Contributing studies

↓ Central analysis team

Imputation



- Goal = get genotypes at untyped positions in the target dataset
- Based on the linkage disequilibrium (LD) between variants
  - "Haplotypes" = group of alleles inherited together
- Reference datasets such as www.1000genomes.org, www.uk10k.org and www.haplotype-reference-consortium.org



TOPMed Imputation Server
Free Next-Generation Genotype Imputation Service

Michigan Imputation Server 2
Free Next-Generation Genotype Imputation Platform

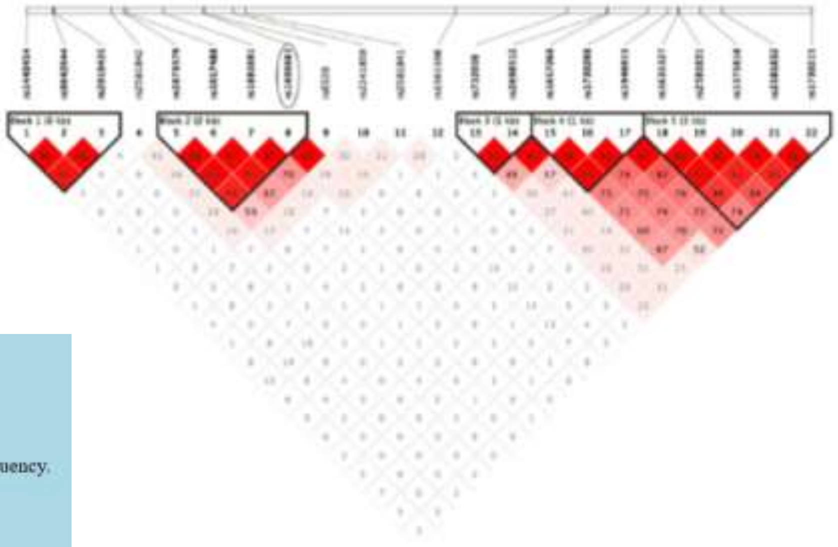
wellcome sanger institute

The Sanger Institute regrets to announce the withdrawal of the Imputation Service. We apologise for any inconvenience this may cause and appreciate your understanding.

Helmholtz Munich Imputation Server (HMIS)
Free Next-Generation Genotype Imputation Service
Empowered by Helmholtz Munich

https://imputationserver.helmholtz-munich.de/index.html#!

Imputation QC



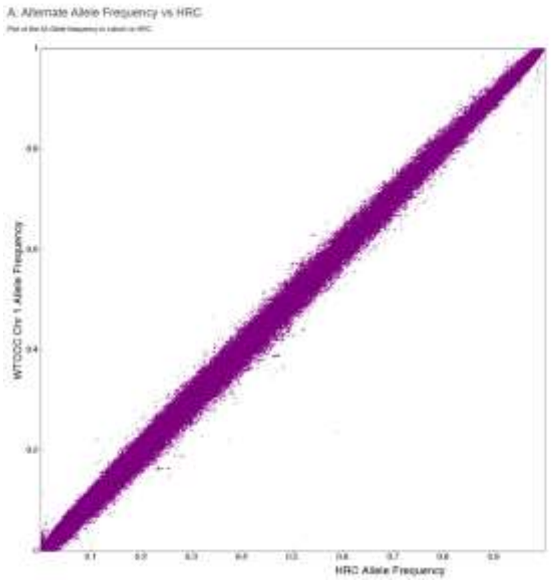
- Imputation preparation and QC checking

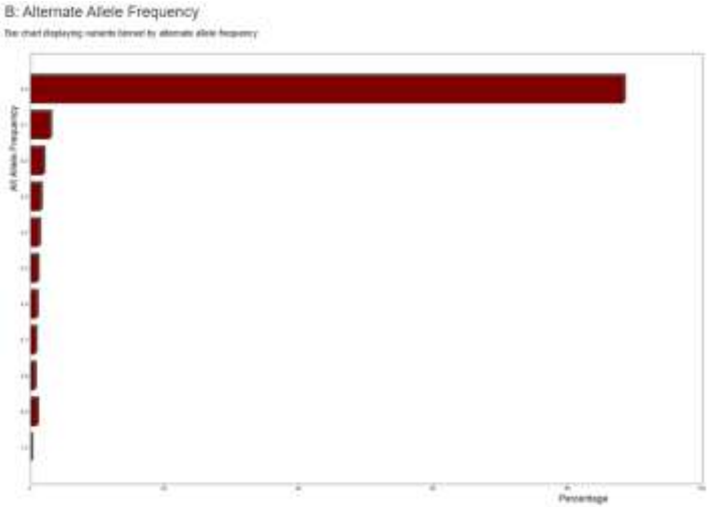
Summary of checks and outputs:

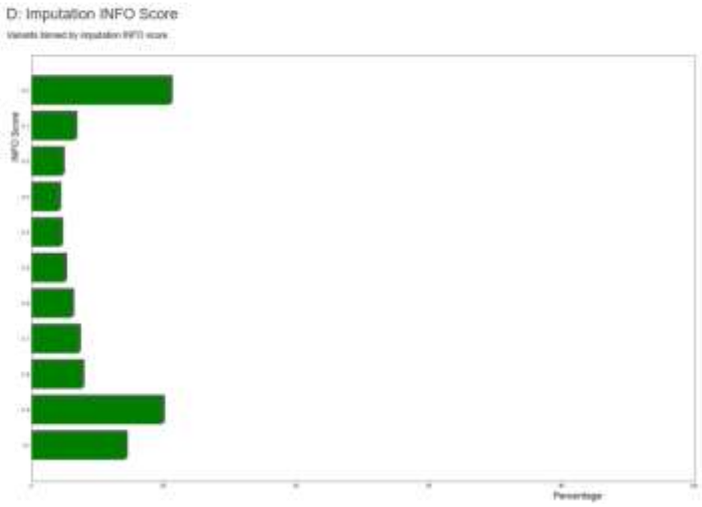
Checks:
Strand, alleles, position, Ref/Alt assignments and frequency differences. In addition to the reference file v4 and above require the plink .bim and (from the plink –freq command) .frq files.
Produces:
A set of plink commands to update or remove SNPs (see below and changes to V4.2.2) based on the checks as well as a file (FreqPlot) of cohort allele frequency vs reference panel allele frequency.
Updates:
Strand, position, ref/alt assignment
Removes:
A/T & G/C SNPs if MAF > 0.4, SNPs with differing alleles, SNPs with > 0.2 allele frequency difference (can be removed/changed in V4.2.2), SNPs not in reference panel

https://www.chg.ox.ac.uk/~wrayner/tools/index.html#Checking

- Post-Imputation data checking







https://www.chg.ox.ac.uk/~wrayner/tools/Post-Imputation.html

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Typical data sharing table format

STUDY TITLE	
General information	
Name of study	
Name of analyst	
Email of analyst	
Study design	population-based, family-based –please give details
Sample information	
Number of cases (females)	
Number of controls (males)	
Ethnic composition	
Possible relatedness issues	are individuals related (how?)
Possible structure issues	mixed population?
Genotyping and imputation information	
Genotyping platform	
Summary of key QC metrics	
# SNPs passed QC	
Imputation method	
Imputation settings	
Reference data used for imputation	including build
Analytical information	
Association analysis method for imputed genotypes	accounting for uncertainty using SNPTEST or other (which?) program, using only genotypes with P(call)>X (which threshold?) as hard calls, using best guess genotypes
Calculated GC lambda (typed SNPs)	
Calculated GC lambda (imputed SNPs)	
Covariates included	PCA, GC, none
Genetic model	

Typical data sharing table format

Column header	Description
SNP	SNP rs number (if unknown, e.g. with some Affymetrix SNPs, report Affy SNP ID)
build	e.g. "36", human genome build used
strand	e.g. "+", human genome strand used
chromosome	chromosome on which SNP resides
position	position of SNP on chromosome in base pairs, based on human genome build used
imputed	"1" for imputed, "0" for directly-typed SNP passing QC
major_allele	e.g. "G", major allele at that SNP, based on control frequency
minor_allele	e.g. "A", minor allele at that SNP, based on control frequency
MAF_controls	e.g. "0.246", minor allele frequency in controls -provide 3 digits to the right of the decimal
OR_allele	e.g. "A", allele to which the OR has been estimated
call_rate	e.g. "0.985", call rate for this SNP across cases and controls -provide 3 digits to the right of the decimal
exact_HWE_cases	exact HWE p value in cases
exact_HWE_controls	exact HWE p value in controls
OR	e.g. "1.097", allelic odds ratio –provide 3 digits to the right of the decimal
lower_95%CI	e.g. "0.874", lower 95% confidence interval of the OR –provide 3 digits to the right of the decimal
upper_95%CI	e.g. "1.267", upper 95% confidence interval of the OR –provide 3 digits to the right of the decimal
additive_p_uncorr	additive model p value, uncorrected for genomic control
additive_p_corr	additive model p value, corrected for genomic control
impute_acc	e.g. "0.98", metric for imputation accuracy (i.e. value for r^2hat or proper_info measures, depending on imputation programme used; if some other measure used, please specify)

Meta-analysis pipeline

Phenotype definition

↓

QC

↓

Imputation (reference, post imputation QC)

↓

Association testing (Correction for covariates and population structure)

↓

Harmonization and QC prior to meta-analysis

↓

Centralized meta-analysis

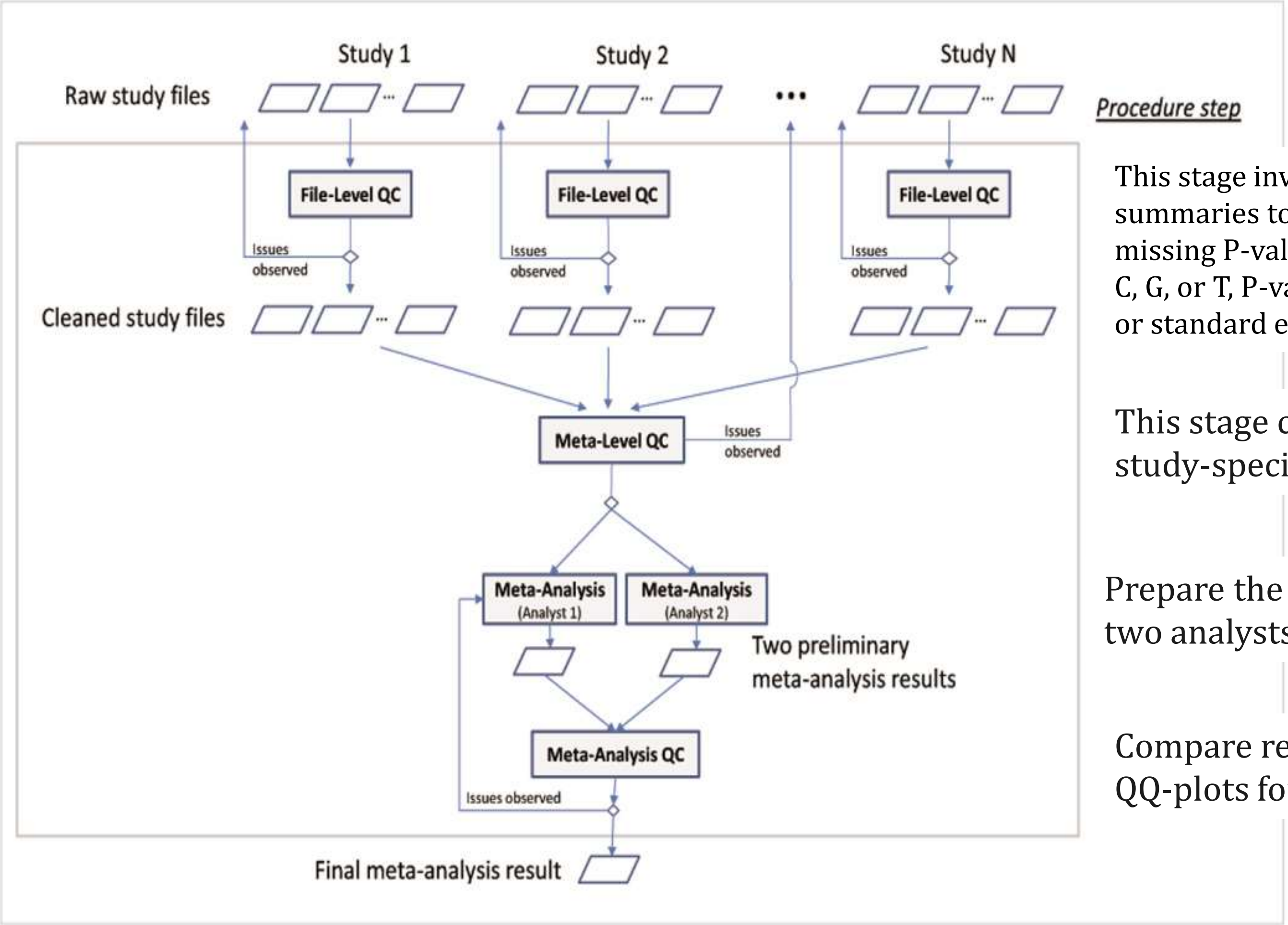
↓

Meta-analysis QC

↓ Contributing studies

↓ Central analysis team

Workflow of the QC and the meta-analysis- EasyQC



This stage involves 'cleaning' (deleting poor quality data) and 'checking' (providing summaries to judge data quality) data. E.g. monomorphic SNPs or SNPs with missing (e.g. missing P-value, beta estimate, or alleles) or nonsensical information (e.g. alleles other than A, C, G, or T, P-values or allele frequencies >1 or <0, or standard errors ≤0, infinite beta estimates or standard errors)

This stage consists in the cross-study comparison of statistics to identify study-specific problems.

Prepare the meta-analysis scripts. For quality control, we recommend that two analysts perform the meta-analysis independently.

Compare results from two meta-analysts (e.g. correlation plots, Manhattan & QQ-plots for further evaluation).

Workflow of the QC and the meta-analysis- EasyQC

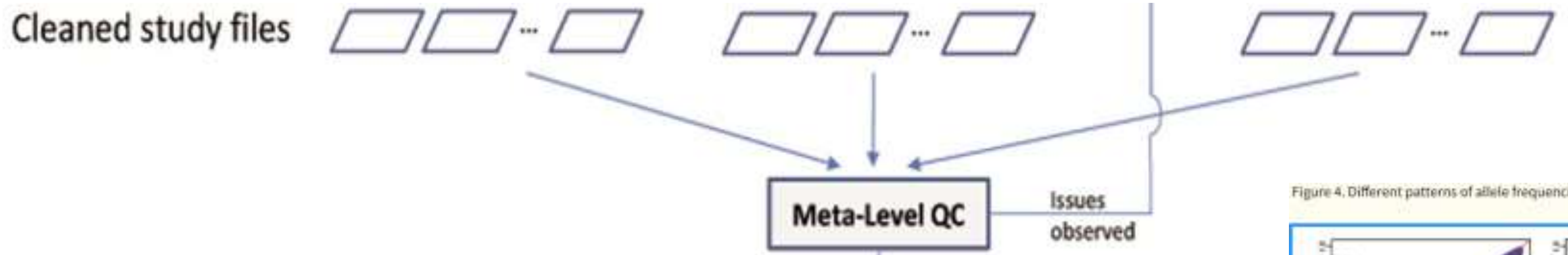


Figure 2. SE-N plots to reveal issues with trait transformations.

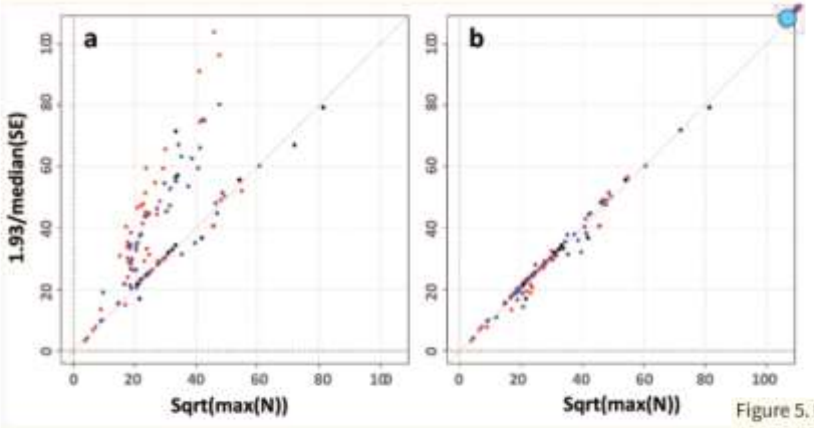


Figure 3. P-Z plot to reveal analytical issues with beta, standard error and P-values.

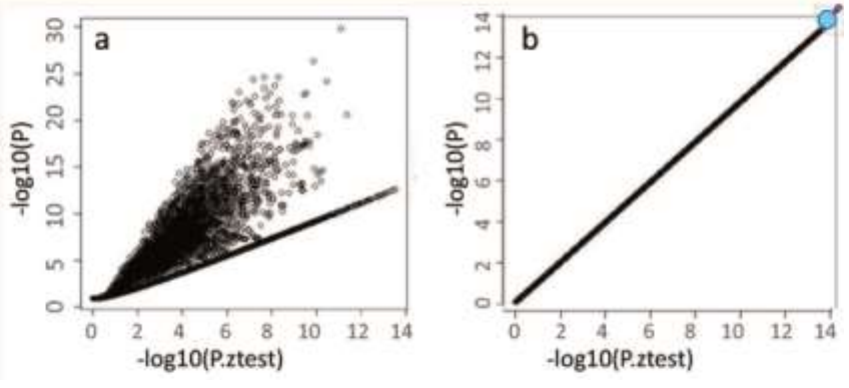


Figure 4. Different patterns of allele frequencies in the EAF plot.

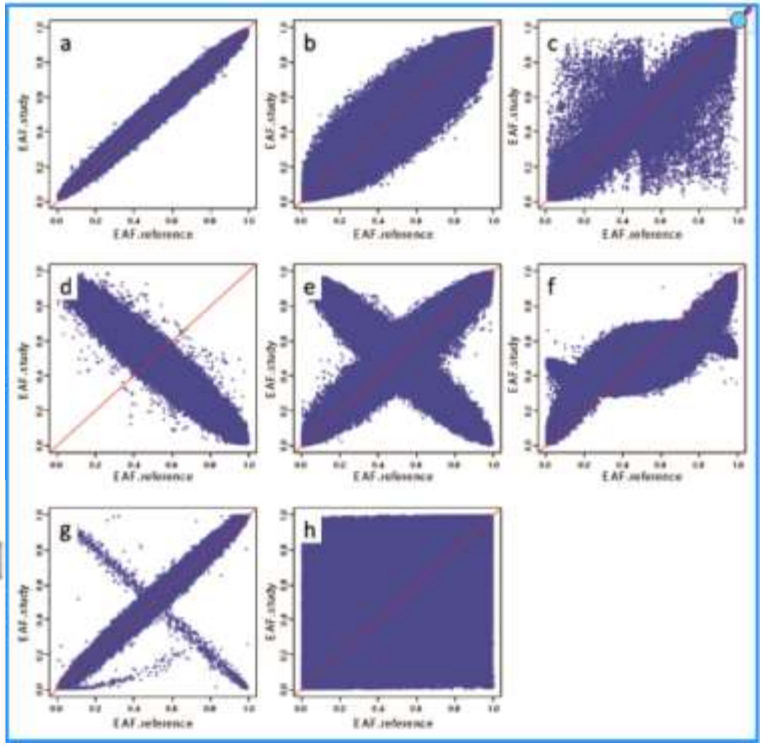
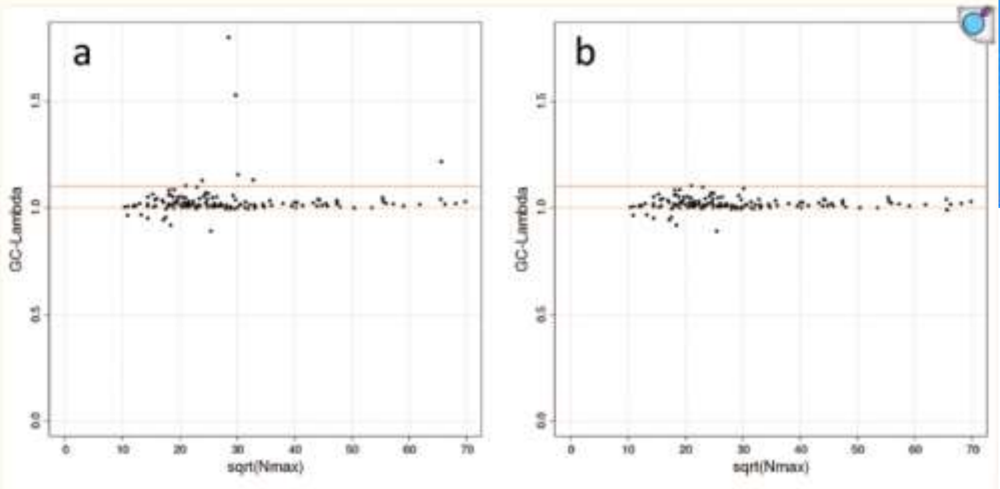


Figure 5. Lambda-N plot to reveal issues with population stratification.



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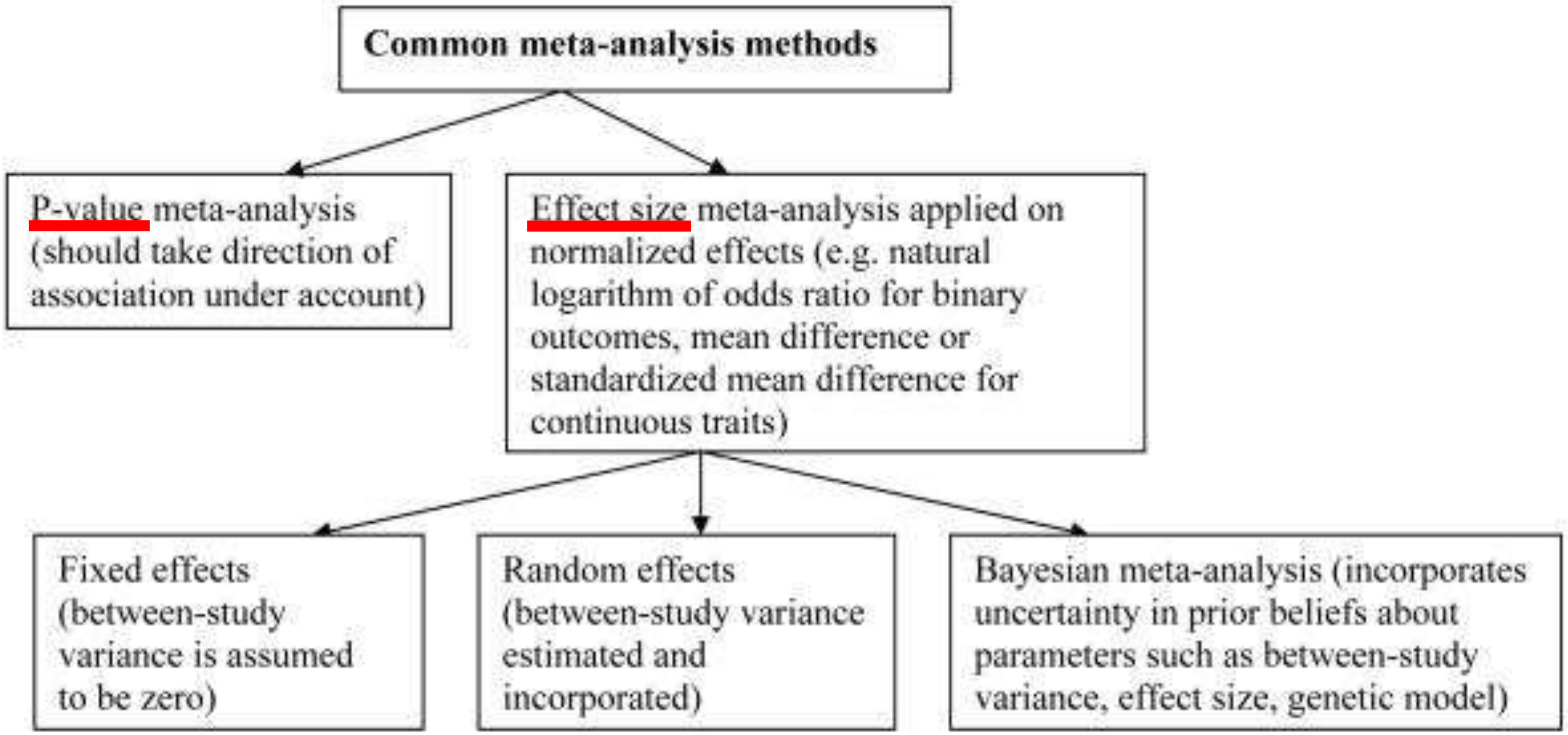
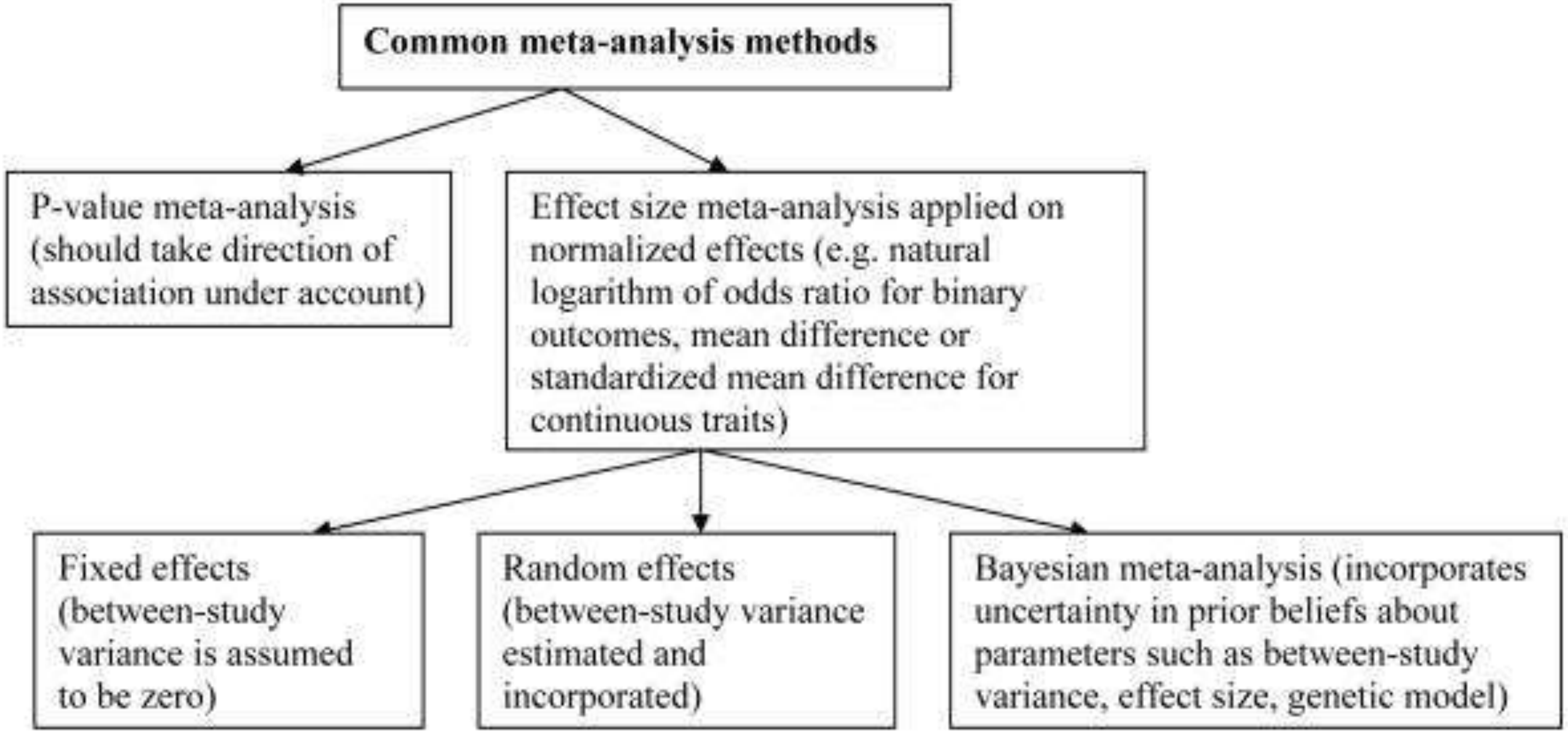
3

Meta-analysis methods

Meta-analysis methods



Meta-analysis methods



The general principle is that studies with more precision should weight more in the calculations than studies with lower precision.

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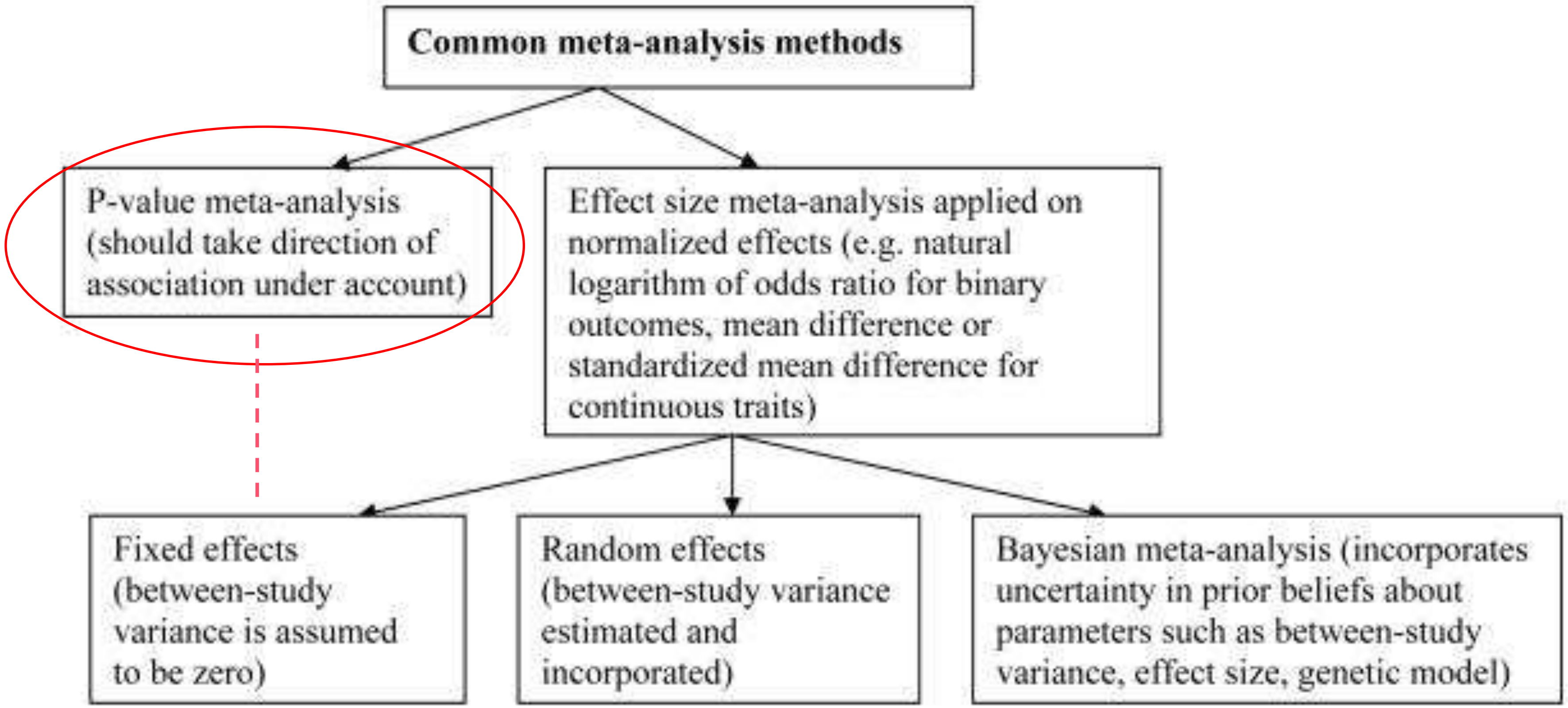
Meta-analysis methods



The major disadvantage of P values meta-analysis is that it cannot provide an overall estimate of effect size. Moreover, between-data-set heterogeneity cannot be assessed, and there is disagreement on the optimal weighting of studies. Furthermore, combining P values may be spurious when the direction of effects in the combined studies is not consistent.
Elements of this approach were used in the Fixed effects model.

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Pooling effect estimates

Phenotype	Analysis	Effect estimate
Case-control	Chi-square test	$OR=e^{\beta}$ $\beta=\ln(OR)$
Case-control	Logistic regression	$OR=e^{\beta}$ $\beta=\ln(OR)$
Quantitative trait	Linear regression	β

3

Meta-analysis methods

3.1

Fixed-effect

Fixed-effect meta-analysis

→ Fixed-effect model assumption: $\beta_1 = \beta_2$ ($\pm\mathcal{E}$)

- Each SNP has a “true effect size” on the trait.
- **This effect is shared by all cohorts (if every study were infinitely large).**
- The observed effect sizes in the different cohorts will be distributed around the “true effect size” with a variance that depends on the precision of the different cohorts (higher weights to the largest studies)

Fixed-effect meta-analysis

Inverse-variance based

- β_i: effect size estimate for study i
- se_i: standard error for study i

Each study's weight is: $\boldsymbol{w_i = \frac{1}{se_i^2}}$

The standard error of the pooled estimate is:
$$SE = \sqrt{\frac{1}{\sum_i w_i}}$$

The combined effect size is the weighted average:
$$\beta = \frac{\sum_i (\beta_i \cdot w_i)}{\sum_i w_i}$$

Z-statistic:
$$Z = \frac{\beta}{SE}$$

P-value: $P = 2 * (1 - \phi(|Z|))$

Sample size based

- N_i: Sample size for study i
- P_i: p-value for study i
- Δ_i: direction of effect for study i

First, convert the p-value and direction into a Z-score:

- $Z_i = \phi^{-1}\left(\frac{P_i}{2}\right) \cdot sign(\Delta_i)$

Then weight each study by:

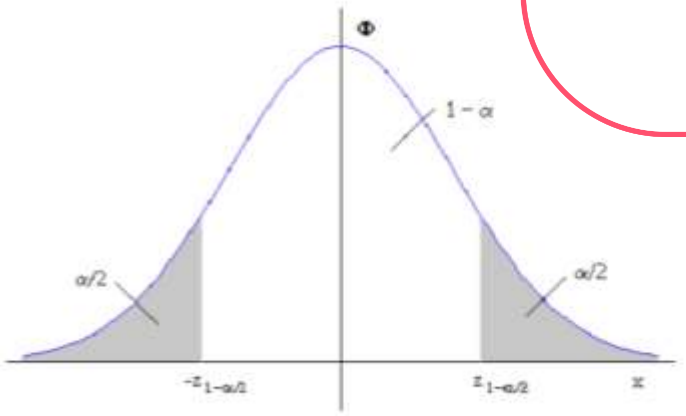
- $\boldsymbol{w_i = \sqrt{N_i}}$

The meta-analysis Z-statistic is a weighted average

- $Z = \frac{\sum_i (Z_i \cdot w_i)}{\sum_i w_i^2}$

P-value:

- $P = 2 * (1 - \phi(|Z|))$



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Correcting for population structure

- Within study correction (before meta-analysis)
  - Use principal components (PCs) or genetic ancestry covariates (e.g., from PCA or mixed-model approaches) in the GWAS regression.
  - Use mixed linear models (MLMs) like: BOLT-LMM, REGENIE, GEMMA, SAIGE. These model both population structure and relatedness.
  - Perform genomic control: Adjust for inflation (λGC) if minor residual inflation remains after PCA correction.
    - Correct test statistics $\chi_i^2 = \left(\frac{\beta_i}{se_i}\right)^2$ by the genomic inflation factor λ $\left(\lambda = \frac{median(\chi_i^2)}{0.456}\right)$
      - χ^2 -> the test statistic for the SNP (square of the Z-statistic).
      - median(χ_i^2) -> The observed median test statistic across all SNPs (genome-wide).
      - 0.456 -> The expected median of a chi-square distribution with 1 degree of freedom under the null hypothesis (no true associations).
        - ≈ 1.0 No inflation.
        - > 1.0 Inflation; test statistics are too large, likely due to population stratification, relatedness, or technical artifacts.
          - Values between >1.05 or >1.10 need attention.
        - < 1.0 Possible overcorrection or conservative test statistics (rare).

- Between studies correction.
  - $X^2 = Z^2 = \frac{\beta^2}{(\lambda * se^2)}$, where λ is the genomic control inflation factor **overall** meta-analyzed association test statistics.

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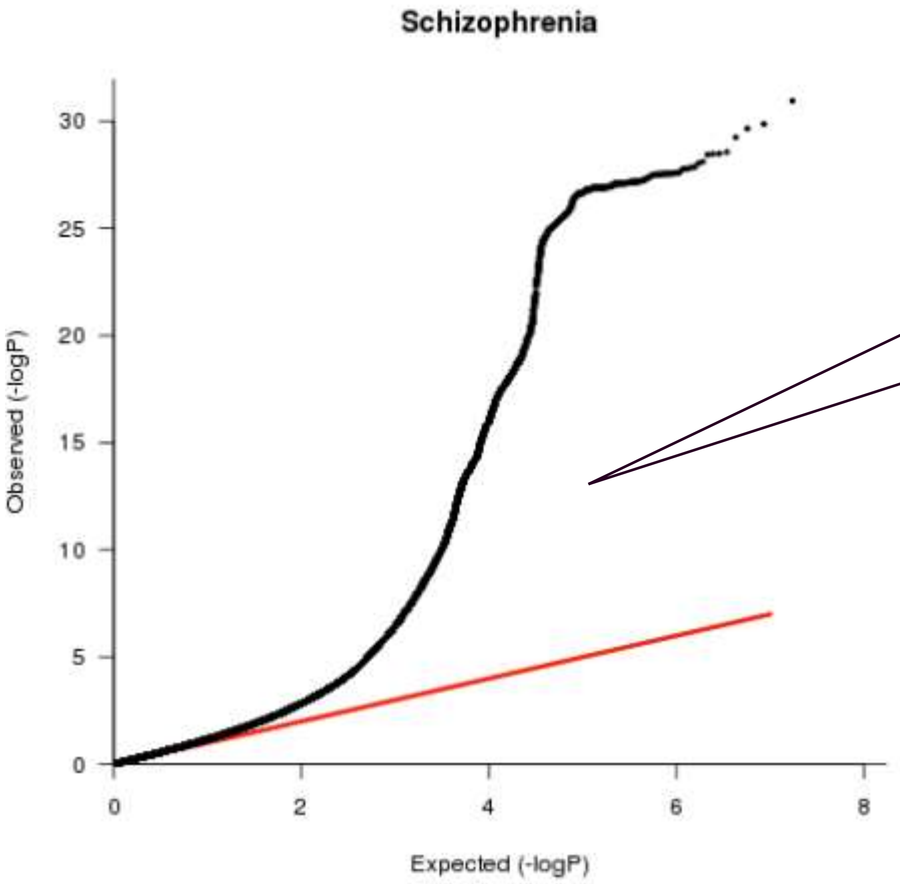
Correcting for population structure

- LD Score Regression Intercept



True polygenic signal
(many small real effects)

+

Confounding (population
structure, cryptic
relatedness, batch effects)

Captures the part of the inflation not explained by polygenic signal → the confounding component.
LDSC intercept adjusts for sample size and polygenicity.

In large, well-powered GWAS meta-analysis of polygenic traits, both GC and LDSR intercept correction leads to power loss, highlighting the need for improved genomic inflation correction methods.

Assessment of heterogeneity

- First step of meta-analysis = assessing heterogeneity across the combined studies

- Statistics:
    - Cochran's Q: is there heterogeneity ?

$$Q = \sum_{i=1}^{N} w_i(\beta_i - \beta_{pooled})^2$$

Q_j for $SNP\ j$ depends on the number of studies for which an allelic effect is reported
Small number of studies: low power

Assessment of heterogeneity

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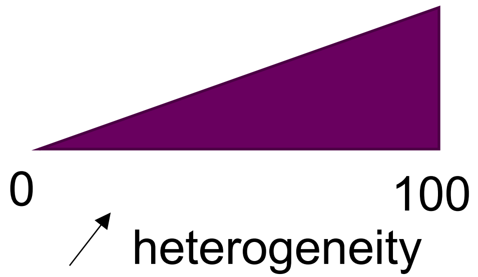
Small number of studies: low power

  - I^2: how much heterogeneity is there ?
    - I_j^2 is more robust to variability in the number of studies included in the meta-analysis for $SNP\ j$
    - $I_j^2 > 50\%$: Large heterogeneity
    - $I_j^2 > 75\%$: Very large heterogeneity



Potential causes for heterogeneity (bias)

- Study-specific bias:
  - Genotyping errors
  - Poor genotype data cleaning
  - Poor imputation
  - Different SNP platforms
  - Population stratification
  - Phenotype misclassification
- Winner’s curse: the originally identified effect size is likely to be overestimated in comparison to its true value

Potential causes for heterogeneity

- Variable LD patterns across studies

  The identified marker is not the causal polymorphism, but has a different LD pattern with the causal polymorphism across different studies.

- Gene-environment interactions with different environmental exposures across populations
- Genuine genetic heterogeneity in effect sizes across different ethnic backgrounds and population-specific effects

Interpreting heterogeneity

- Heterogeneity may represent genuine differences in genetic effects across different populations and different biological settings (**truly informative heterogeneity**)
- Informative heterogeneity may reveal interesting facts about biology, e.g. the mechanism through which the variant is acting on disease risk.
- Recognizing the potential for heterogeneity can rescue associations from being discarded as replication failures.

Interpreting heterogeneity

- Heterogeneity may represent genuine differences in genetic effects across different populations and different biological settings (**truly informative heterogeneity**)
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- Recognizing the potential for heterogeneity can rescue associations from being discarded as replication failures.
- Example of informative heterogeneity: relationship between the obesity associated (FTO) gene and T2D.
    - Overall the evidence suggests that FTO affects weight, which in turn increases risk of T2D.
    - As a consequence of being on the same pathway, FTO showed an association with T2D in population-based studies but failed to replicate in studies that controlled for weight by only recruiting lean subjects.

3

Meta-analysis methods

3.2

Random-effect

Random-effect models

→ Random-effect model assumption: $\beta_1 \neq \beta_2$

- The true effect for a SNP varies between cohorts.
- The studies included in the meta-analysis are assumed to be a random sample reflecting the distribution of true effects

The weight of each study incorporates the between-study variance of heterogeneity τ^2 which is used to estimate allelic effect in each study.

Random-effect models

- Used if large differences across samples
- Same scale is used across samples
- Number of samples should be sufficiently large

$$\tau^2 = \frac{\boldsymbol{Q} - (k-1)}{\sum_{i-1}^{N} w_i - \frac{\sum w_i^2}{\sum w_i}}$$

$$w_i = \frac{1}{se_i^2} \rightarrow w_i^* = \frac{1}{(\tau^2 + se^2)}$$

- Random effect models are more conservative (larger se)

Specialized software for GWAS meta-analysis

GWAMA
Genome-Wide Association Meta-Analysis
www.well.ox.ac.uk/gwama/contact.shtml

METAL
Meta-Analysis Helper
www.sph.umich.edu/csg/abecasis/metal

META
www.stats.ox.ac.uk/ jsliu/meta.html

MetABEL
R package part of GenABEL, an R library for GWAS analysis
www.genabel.org/packages/MetABEL

METASOFT
Han and Eskin’s Random Effects model, Binary Effects model
http://genetics.cs.ucla.edu/meta
Ioanna Tachmazidou

Comparison of software

	METAL	GWAMA	MetABEL	PLINK	R packages
Ability to process files from GWAS analysis tools; software used	No	Yes; SNPTEST, PLINK	Yes; ABEL	Yes; PLINK	No
Fixed effects implemented?	Yes	Yes	Yes	Yes	Yes
Random effects implemented?	No	Yes	No	No	Yes
Heterogeneity metrics generated	Q, I^2	Q, I^2	Q, I^2	Q, I^2	Q, I^2
Graphical illustration of meta-analysis results	No	Manhattan and QQ plots	Forest plots	No	Yes

GWAS, genome-wide association study.

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Magi and Morris, BMC Bioinformatics, 2010

3

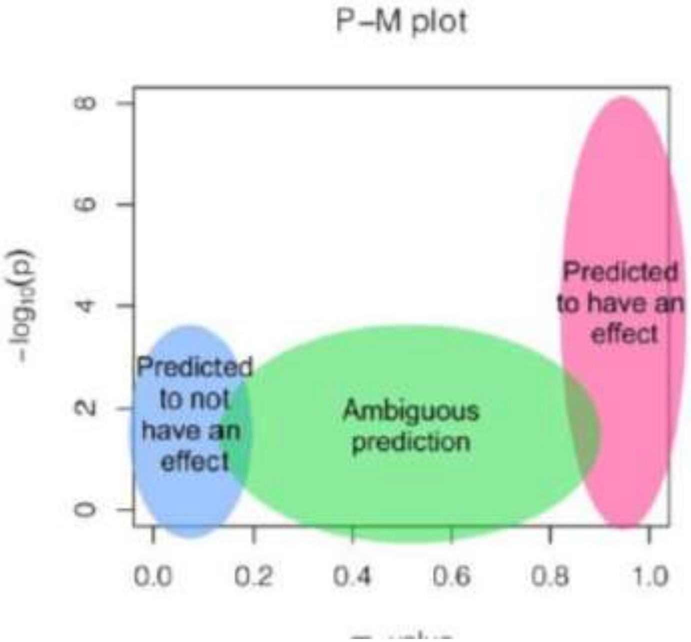
Meta-analysis methods

3.3

Bayesian

Bayesian meta-analysis: Binary effect model

- Based on the posterior probability that the effect exists in each study (m-value)
- Estimated using cross-study information via Markov Chain Monte Carlo (MCMC) method.
- Segregates the studies predicted to have an effect, the studies predicted to not have an effect, and the underpowered ones



- If m-value > 0.9, the study is predicted to have an effect
- If m-value < 0.1, the study is predicted not to have an effect

- Binary effect test statistics

$$S_{BE} = \frac{\sum m_i \sqrt{W_i} Z_i}{\sqrt{\sum m_i^2 W_i}}$$

Where $Z_i = \frac{\beta_i}{se_i}$ and $\sqrt{W_i} = \sqrt{Ni}$

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Han and Eskin, PLoS Genetics 2012

3

Meta-analysis methods

3.4

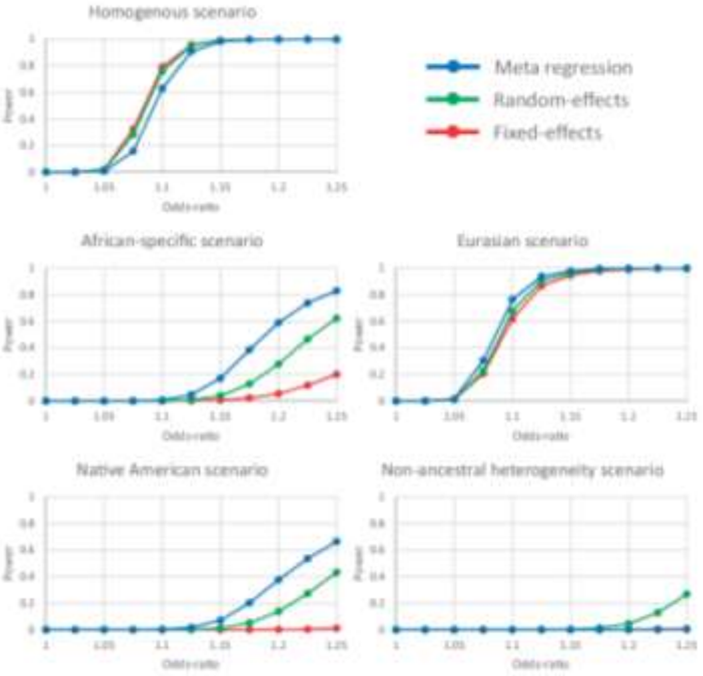
Trans-ancestry meta-analyses

Meta-regression

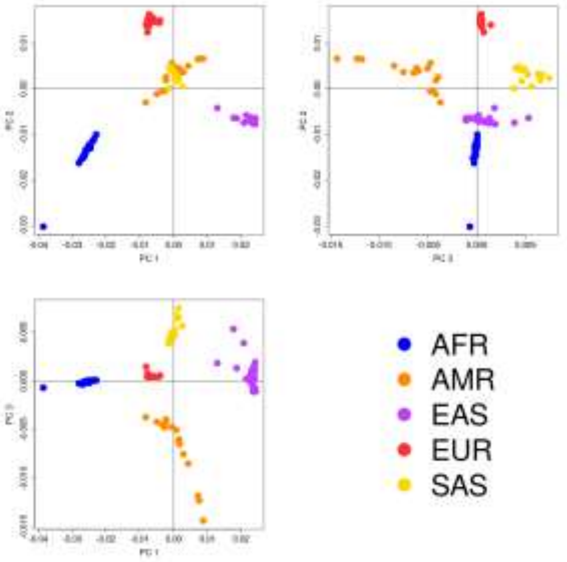
- Magi et al. 2017 & Suzuki et al. 2024

  → As we venture into larger and more genetically diverse investigations, the challenges of extending observations across ancestry groups become more prominent.
  → Take into account heterogeneity related to ancestry
  → Integrates axes of genetic variations (PCs from PCA) in a linear regression framework (models the genetic diversity as continuous and multidimensional)

  → MR-MEGA is a specialized software



$$E\left(b_{kj}\right) = \alpha_j + \sum_{t=1}^{T} \beta_{tj} x_{kt}$$



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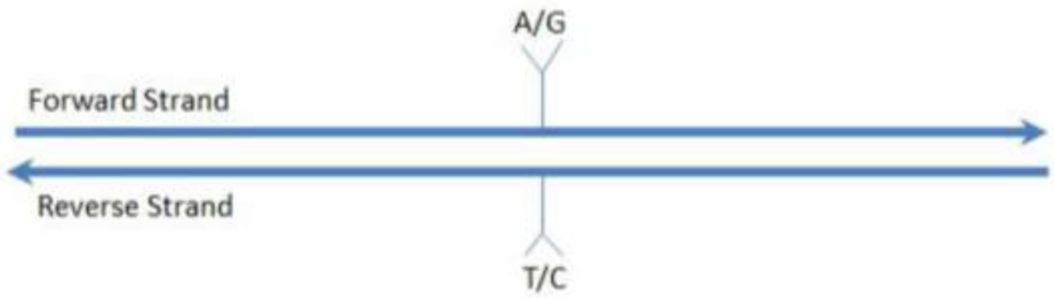
Considerations

Remapping genome build and strand flipping

- Ensuring all data are aligned to the same strand on the same genome build

  → Ensure the same risk allele across studies (effect sizes are associated to an allele)
  → Usually the forward or positive (+) strand

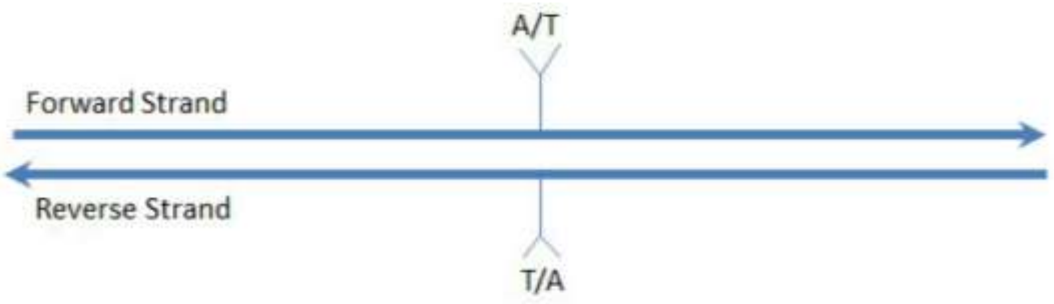
- For most SNPs, the strand can be identified by the alleles



- SNP is A/G on the forward strand and the T/C on the reverse, thereby uniquely identifying the strand

Alleles are not the full story

- For some SNPs, the strand cannot be determined using the alleles



- In this case, the alleles on either strand are A/T
- This is the same for G/C SNPs

Using allelic frequency to determine the strand

- Assume SNP is A/T, the Minor Allele is A with a frequency (MAF) of 30%.
- A second study with the SNP listed as Minor Allele T with a frequency of 32% is likely on the opposite strand.
- This is not conclusive as SNPs vary in frequency between populations.
- A/T or G/C SNPs with a frequency near 50% are particularly difficult to determine using frequency.

Study alignment and error trapping

Example of alignment of allelic effects and error trapping or a single SNP in a meta-analysis of five studies of a dichotomous phenotype

Study	Reported strand	Effect allele[1]	Other allele	RAF	Odds ratio (95% confidence interval)	Aligned allelic effect (standard error)	Comment
1	+	A	G	0.12	1.12 (1.07-1.16)	0.11 (0.02)	Allele A taken as reference effect allele.
2	+	G	A	0.85	0.92 (0.87-0.98)	0.08 (0.03)	Effect aligned to allele A.
3	-	T	C	0.12	1.06 (1.02-1.10)	0.06 (0.02)	Effect aligned to allele A on + strand.
4	+	T	C	0.13	1.07 (0.99-1.16)	0.07 (0.04)	Effect aligned to allele A on + strand. Strand error reported to log file.
5	+	A	G	0.87	0.95 (0.90-1.01)	-0.05 (0.03)	Large discrepancy in EAF reported to log file.

[1] Effects are aligned to the reference allele in the first study. Errors in the reported strand are recorded in the log file together with warnings regarding potential discrepancies in reported data between studies, for example the aligned reference allele frequency (RAF).

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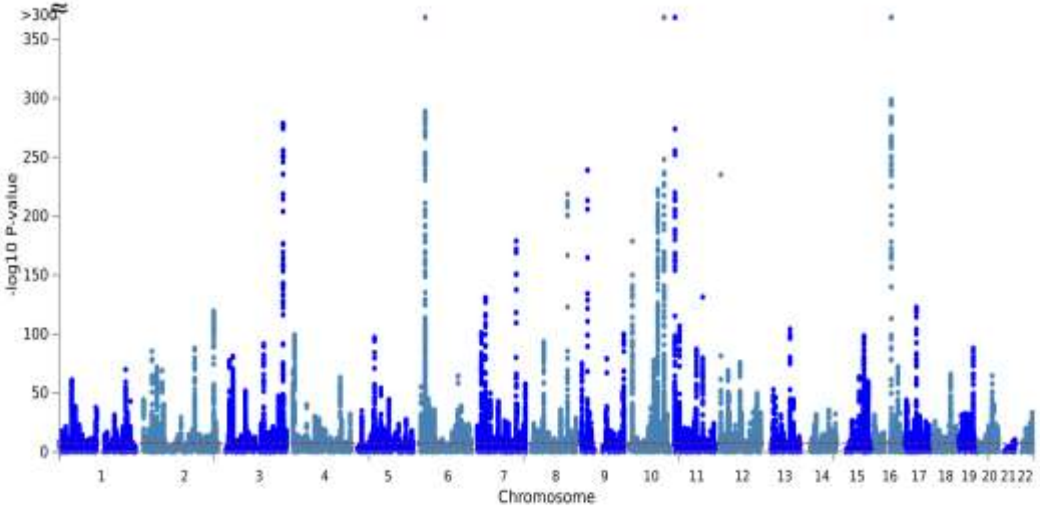
Magi and Morris, BMC Bioinformatics, 2010

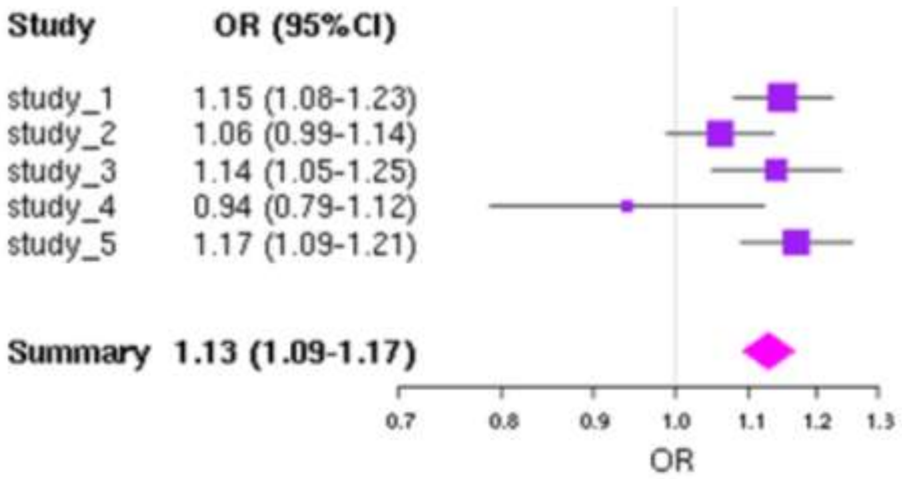
5

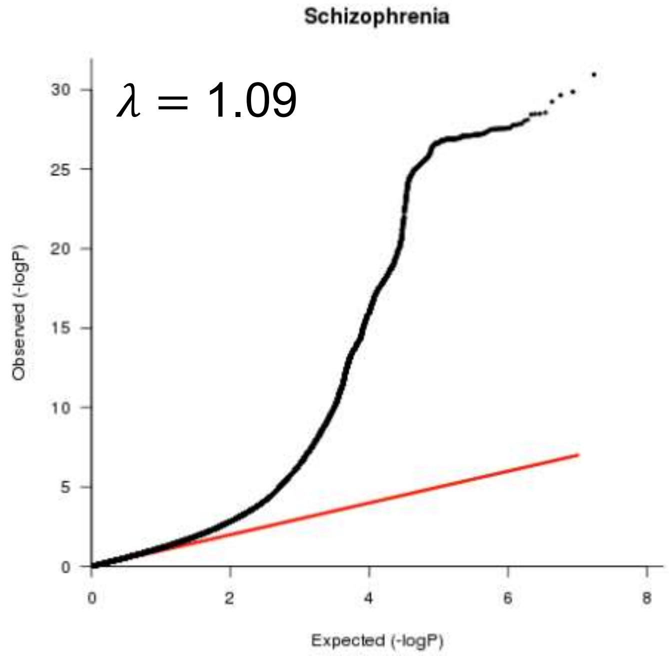
Meta-analysis results

Visualization

- Classical tools: QQ-plots, Manhattan plots
- Forest plots: study-specific and overall effect







Interpretation

- No between-study heterogeneity:

  → Fixed effect = random effect calculations
  → Identical point estimates and confidence intervals

- Increasing between-study heterogeneity

  Random effects summary estimates have larger variance (wider confidence intervals)
  → Usually less prominent statistical significance

- Most meta-analysts would typically run both models
- If the between-study heterogeneity comes from ancestry, then MR-MEGA is used as well.
- In all situations: statistically significant associations need **replication**

Interpretation

- Effect sizes should be estimated with precision
  - The analytical model used, e.g. genetic model specification, may affect the magnitude of the effect size
  - Winner's curse effect: significant associations are likely to have observed effect sizes that are inflated compared with the true effect size

- Is a newly discovered genetic variant the causal variant or simply linked to it?
  - Hard to answer
  - Even large-scale GWA meta-analyses require extensive **fine-mapping, targeted resequencing and functional follow-up** experiments before the truly causal variants can be identified.

- Discovery of a genetic locus has important implications on its own

  → May highlight some interesting biological pathway and may give some insights into developing new therapeutics.

Replication

- Prioritize interesting signals for follow-up and replication.
- Follow-up sample sets should be adequately powered to detect the association.

  → The replication stage could be a large meta-analysis itself.
- Issues of set-up, information aggregation, estimation of heterogeneity and summary effects in a replication effort are similar to those described for discovery meta-analysis.
- Meta-analyze the discovery with the replication data to capture the totality of the evidence.
- Review the literature and bioinformatic databases to identify candidate variants both for the particular trait under study and for associated traits.
- Replication of previously published hits confirms previous publications and gives validity to the meta-analysis.

Stages of a meta-analysis

- Sensitivity analysis
  - Does the effect extend over a chromosomal region or is it confined to one variant ?
  - Do the results depend critically on a single study ? Why ?
  - Is the effect stronger under a recessive or dominant genetic model ?
  - Is there large heterogeneity at a locus ?

- Secondary analysis
  - Assess the importance of some variants adjusted for others, in order to see if the two sets act independently
  - Adjust for phenotypes that lie on potential causal pathways
  - Is the signal driven or stronger in males or females ?
  - Is the signal driven by some specific covariates ?

Summary

- Genetic effects of **common variants** associated with complex diseases are mostly **moderate/small** and require very large sample sizes to identify with certainty.
- Meta-analysis of genome-wide data can improve the power for detecting and validating such associations.
- Meta-analysis of data from studies using different platforms is enhanced by the use of **imputed** genotype data.
- Careful **collection and quality-checking** of information is essential to avoid errors.
- A wide array of methods may be used, including fixed effects, random effects, Bayesian and multi-ancestry meta-analysis and they have particular advantages and disadvantages.
- Application of the meta-analysis methodology in genome-wide data has been successful in identifying more disease-related genes for various conditions.

Thank you.

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